

The exam paper includes 16 pages

Exam instructions:

- All MCQs have to be answered in bubble sheet.
- Make sure that your exam form is similar to bubble sheet model.
- Write answers of MCQ in bubble sheet by blue pen only.
- Double answer of any MCQ will be considered wrong.

1) MCQ (select the best answer) (60 questions, 0.5 mark for each question)

1. The following is recognized cause of dilated cardiomyopathy:

- Systemic sclerosis. X
- Polymyositis.
- Hemochromatosis X
- SLE.

2. Which type of cardiomyopathy can be caused by amyloidosis?

- Infiltrative
- congested.
- Dilated
- Hypertrophic.

3. Regarding asbestos related illnesses

- Family members of asbestos workers are not at increased risk of asbestos related cancers
- Mesothelioma but not lung carcinoma is a common cancer related to asbestos exposure
- Serpentine asbestos fiber types are the most pathogenic.
- Concomitant smoking is not an increased risk for carcinoma related to asbestos exposure. X

4. Siderosis is due to ?

- Iron and iron oxides
- Tin oxide
- Barium sulfate
- Antimony salts

5. Which of the following tests is least relevant in sarcoidosis

- 24-hour urinary excretion of calcium ✓
- 24-hour proteinuria
- Slit-lamp and fundus examination
- Serum angiotensin converting enzyme

6. Which of the following is related to treatment of childhood tuberculosis?

- Corticosteroids is contraindicated in treatment of TB
- Rifampin for 9 months for isoniazid resistant
- Isoniazid for 9 months for isoniazid susceptible
- Pyrazinamide for isoniazid rifampin resistant

7. What would be an appropriate first-line treatment for the management of

heart disease?
A. indomethacin administration
prostaglandin E1 infusion
nitric oxide
Surfactant

→ 8. A full-term newborn delivered via normal spontaneous vaginal delivery with Apgars of 9 at 1 minute and 9 at 5 minutes is noted to have a grade 2/6 systolic murmur at the left sternal border on day 1 of life. Upon reexamination on day 4 of life, the neonate is found to have poor peripheral pulses, hypotension, and shock. What is the least likely cause of shock in this neonate?

- ☒ hypoglycemia
- ☐ sepsis
- ☐ perinatal asphyxia
- ☐ cardiogenic shock

9. The most common cause of sudden death during competitive exercise is?

- ☐ Postoperative congenital heart disease
- ☒ Hypertrophic cardiomyopathy
- ☐ Marfan syndrome
- ☐ Aortic valve stenosis

10. Two weeks after having a nonspecific upper respiratory tract infection, a previously healthy 4-year-old boy is noted to have a respiratory rate of 40 breaths per min, a heart rate of 140 beats per min. His heart rate is occasionally irregular and a gallop rhythm is noted. A 1/6 blowing systolic murmur is heard at the apex. He has moderate hepatomegaly. Which of the following is the MOST likely diagnosis?

- ☐ Acute rheumatic fever
- ☐ Infective endocarditis
- ☒ Viral myocarditis
- ☐ Paroxysmal atrial tachycardia

RR 40
HR 140

11. A sixty-year-old man, heavy smoker complaining of single, central invasive lung mass. Histological examination revealed small, non-cohesive tumor cells, indistinct cytoplasm and hyperchromatic nuclei. The most likely diagnosis is;

- ☐ Squamous cell carcinoma
- ☒ Small cell carcinoma
- ☐ Adenocarcinoma
- ☐ Metastatic carcinoma

Smoking - central

12. Which of the following's lung tumor has a tendency to undergo necrosis and cavitations?

- ☐ Adenocarcinoma
- ☐ Bronchioalveolar carcinoma
- ☐ Carcinoid
- ☒ Squamous cell carcinoma

13. Which of the following may complicate a healed myocardial infarction?

- ☒ Myocardial aneurysm
- ☐ Ventricular fibrillation
- ☐ Acute pulmonary edema
- ☐ Acute left-sided heart failure.

Long-acting beta two agonist
following?

- A- Delay up regulation
- B- Enhance down regulation
- ☒ C- Prevent occurrence
- D- Decrease the effect

15. -Regarding the following

- A- Erythema
- ☒ B- Pharyngitis
- C- Cough
- D- Dyspnea

16. W

14. Long-acting beta two agonists should be used with inhaled corticosteroids due to which of the following?

- ☐ Delay up regulation of receptors.
- ☒ Enhance down regulation of receptors.
- ☐ Prevent occurrence of tolerance.
- ☐ Decrease the hypokalemic effect.

15. -Regarding the drug interaction of theophylline. The half-life of theophylline is decreased by which of the following?

- ☐ Erythromycin ✗
- ☒ Phenobarbitone
- ☐ Quinolones
- ☐ Cimetidine ✗

16. Which one of the following, beta two agonists can be taken once per day?

- ☐ Albuterol
- ☐ Salmeterol
- ☒ Terbutaline
- ☐ Indacaterol

17. The left lung is supplied by:

- ☐ one bronchial artery
- ☒ two bronchial arteries
- ☐ one pulmonary artery
- ☐ two pulmonary arteries

18. The venous drainage of the right lung is:

- ☐ superior vena cava
- ☒ azygos vein
- ☐ superior hemiazygos vein.
- ☐ Inferior hemiazygos vein

19. The heart is located at:

- ☐ Anterior Mediastinum
- ☒ Posterior Mediastinum
- ☐ Superior Mediastinum
- ☐ Middle Mediastinum

20. A 75-year-old woman with a history of chronic congestive heart failure undergoes cardiac catheterization to determine her extent of cardiac dysfunction. During estimation of her systolic function, in what phase of the cardiac cycle should her left ventricular volume unchanged?

- ☐ Rapid filling.
- ☒ Isovolumetric contraction
- ☐ Ventricular ejection
- ☐ Atrial systole

21. Why the pulmonary artery BP is lower than the systemic arterial BP?

- The output of the right ventricle is less than that of the left ventricle.
The left ventricular wall is thicker than that of the right ventricle. d
The aorta is more elastic than the pulmonary artery.
The pulmonary resistance is less than the systemic resistance.

22. Which of the following is true regarding pulmonary circulation?

- It is a low-pressure system that allows for improved gas exchange.
It is the larger of the two circulatory systems. x
The system functions with an increased arterial pressure to circulate through the distal parts of the body.
It consists of the left side of the heart, the aorta, and its branches. x

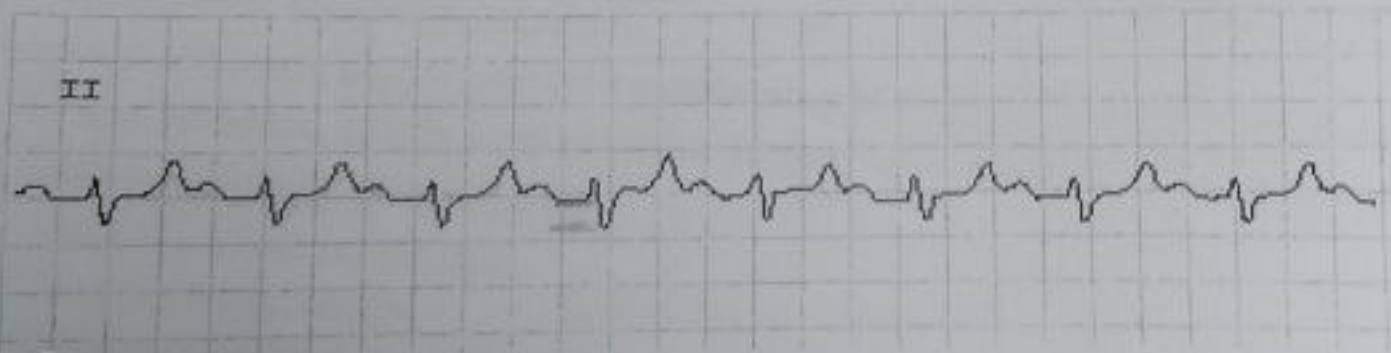
23. Male patient 50 years old, is known to be hypertensive presented to emergency room complaining of Sudden onset of severe chest pain with 'tearing' or 'ripping' in quality, pain radiates to the back and legs. What is the most probable diagnosis?

- Coarctation of the aorta
Acute myocardial infarction
Aortic dissection.
Acute pulmonary embolism

→ 24. Angina equivalents are presented by?

- Dyspnoea
Palpitation
Shock
Stitching pain.

25. Please review the following rhythm strip, what is the main finding?



- Sinus rhythm with first-degree AVB
Junctional rhythm
Accelerated idioventricular
Normal sinus rhythm

26. What is the most common type of arrhythmia in patients with mitral stenosis?

- Ventricular tachycardia
Atrial fibrillation
Ventricular premature beats
Torsad de points

27. The most common cause of pulmonary stenosis is:
☒ Bacterial endocarditis
☐ Rheumatic endocarditis
☐ Congenital
☐ Carcinoid syndrome

28. 47 What is the character of murmur in case of chronic mitral valve regurgitation.
☒ Mid systolic click and late systolic murmur heard at apex
☐ Mid diastolic click and late diastolic murmur heard at apex
☐ Holodiastolic murmur at apex
☐ Holosystolic murmur at apex propagated to axilla

MA
holo systolic

29. A 30 year old IV drug user presents with unexplained fever for 4 weeks associated with anorexia, weight loss, on examination he has hemorrhagic lines under nail with painful nodule at the figure tips, cardiac auscultation revealed new onset systolic murmur over the apex shortness of breath, night sweats and a fever. On examination, you find that he has splinter hemorrhages. What is the appropriate investigation to confirm the diagnosis?

- ☒ Chest x-ray
☐ Blood culture
☐ Abdominal ultrasound
☐ ECG

splinter hgr
Osler's
New syst. mur

30. which of the following condition is associated with high risk of developing infective endocarditis

- ☒ Mitral valve prolapse without regurgitation
☐ The presence of a prosthetic heart valve.
☐ History of rheumatic fever without valvular defects
☐ Coronary intervention with coronary stenting

31. To diagnose a patient with rheumatic fever, The patient should have at least how many criteria:

- ☒ One major, two minor
☐ Two major, one minor
☐ One major, one minor
☐ Two major, two minor

(Case) 1 + 2
2 + 1

32. Rheumatic chorea is characterized by?

- ☒ More common in adolescent male. X
☒ Symptoms continue during sleep. X
☐ May occur late after streptococcal infection up to 3 months.
☐ Seen approximately in all patients X

33. What is the chamber with the arrow in the shown picture?

- ☒ Left ventricle
☐ Left atrium
☐ Right atrium
☐ Right ventricle.



34. The most common cardiac tumor is:

- ☒ Atrial myxoma
- ☐ Rhabdomyoma
- ☐ Papillary fibroelastoma
- ☐ Fibroma

35. which of the following is considered positive for coronary insufficiency of exercise ECG test:

- ☒ High blood pressure response to exercise
- ☐ Horizontal ST segment depression
- ☐ Fatigue
- ☐ Failure to increase heart rate to exercise.

36. Hoarseness of voice could be a presenting symptom in patient with:

- ☒ Aortic aneurysm
- ☐ Dissection of the aorta
- ☐ Coarctation of the aorta
- ☐ Mitral regurgitation

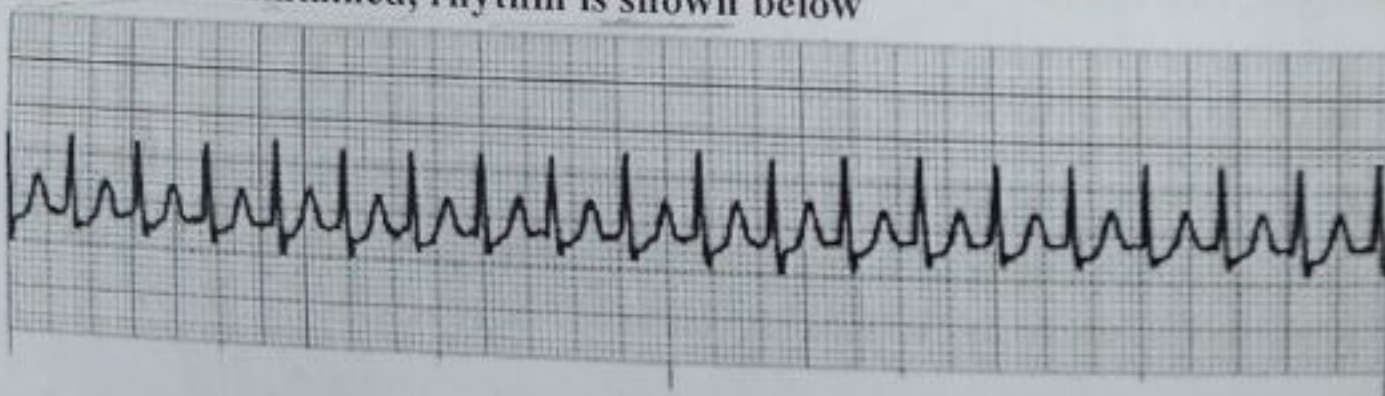
37. A 25 year old young man presented with fever, stitching chest pain and auscultatory friction sound over the precordium. With patient decubitus, what is the most probable diagnosis

- ☒ Acute coronary syndrome
- ☐ Acute pericarditis
- ☐ Acute pulmonary embolism
- ☐ Aortic dissection

38. Narrowing of the aorta just beyond the left subclavian artery at the insertion of ductus arteriosus is seen in case of:

- ☐ Aortic aneurysm
- ☐ Aortic dissection
- ☒ Coarctation of the aorta
- ☐ Pulmonary hypertension

39. A 30 year old young lady came to emergency room with sudden onset of rapid palpitation, her blood pressure was maintained, rhythm is shown below



The best first line of management in this situation is

- ☒ Vagal maneuvers
- ☐ IV Ca channel blocker
- ☐ IV Amiodaron
- ☐ IV Digitalis

40. Smoking predisposes to which of the following tumors?

- ☐ Choriocarcinoma
- ☐ Yolk sac tumor
- ☐ Embryonal carcinoma
- ☒ Bronchogenic carcinoma

41. The most common complication post cardiac surgery

- ☐ Bleeding
- ☐ Infection
- ☐ Stroke
- ☒ Arrhythmias

42. A 65-year-old male, with history of asbestos exposure presented with chest pain and fever. His ESR 140 and a chest X-ray showed right-sided pleural thickening with effusion. Which of the following is the most possible diagnosis?

- ☒ Mesothelioma.
- ☐ Interstitial lung disease.
- ☐ Pneumonia.
- ☐ Bronchial asthma.

65
asbestos
ESR ↑

→ 43. Which one of the following is not true regarding lung tumors?

- ☐ 97% of lung tumors are malignant. ✓
- ☐ Bronchogenic carcinomas often present late when local invasion and metastases are already present.
- ☐ Lung cancer is by far the number one cancer killer. ✓
- ☒ 50% are due to smoking.

90

44. What is the most common cause of transudative pleural effusion?

- ☐ Trapped lung.
- ☒ Heart failure.
- ☐ Nephrotic syndrome.
- ☐ Peritoneal dialysis.

45. A 70-year-old female, her husband is a constructor presented with dyspnea and chest pain. Chest X-ray showed small RT sided pleural effusion and pleural nodules. Aspiration of pleural fluid revealed exudative hemorrhagic pleural effusion. Which of the following is the most diagnostic next step?

- ☒ Transthoracic ultrasound guided pleural biopsy.
- ☐ Thoracoscopic guided pleural biopsy.
- ☐ Abrams pleural biopsy.
- ☐ CT guided pleural biopsy.

46. Theophylline use is can be limited by which of the following?

- ☐ Need to constantly monitor serum levels.
- ☐ Has weak bronchodilator effect.
- ☒ A narrow therapeutic index.
- ☐ Numerous GIT side effects.

47. Which of the following is not normally associated with bronchial asthma?

- ☐ Decreased diffusion capacity.
- ☐ Increased residual volume. RV
- ☒ Cyanosis at rest.

D. Hyperinflation.

48. What drug therapy should be available to all asthma patients (intermittent and persistent)?

- ☒ Low dose inhaled corticosteroids (ICS).
- ☐ PRN short acting β_2 -agonist (on demand SABA).
- ☐ Long acting β_2 -agonist (LABA).
- ☐ Leukotriene antagonist.

49. Which of the following is not a characteristic of bronchial asthma?

- ☒ Increase in IgG immunoglobulins.
- ☒ Airway hyperresponsiveness.
- ☒ Infiltration of eosinophils into the airways.
- ☒ Increased mucus production.

50. Thymoma is.....

- ☒ Anterior mediastinal lesion.
- ☐ Posterior mediastinal lesion.
- ☐ Middle mediastinal lesion.
- ☐ Superior mediastinal lesion.

51. A 54 year old male, recently bedridden due to fracture femur. He presented with shortness of breath and fever. Which of the following is highly sensitive for ruling out (exclusion) of PE and is used for patients whose probability for PE is unlikely?

- ☒ V/Q scanning.
- ☐ D-dimer testing.
- ☐ CT pulmonary angiography.
- ☐ Duplex ultrasonography.

52. Which of the following are the first line antituberculous drugs?

- ☒ Pyrazinamide, Isoniazid, Rifampicin, Ethambutol.
- ☐ Pyrazinamide, Isoniazid, Ethionamide, Ethambutol.
- ☐ Kanamycin, Isoniazid, Capreomycin, Ethambutol.
- ☐ Kanamycin, Azithromycin, Capreomycin, Ethambutol.

53. A firefighter who was involved in extinguishing a house fire is being treated for smoke inhalation. He develops severe hypoxia 48 hours after the incident, requiring intubation and mechanical ventilation. Which of the following conditions has he most likely developed?

- ☒ ARDS.
- ☐ Atelectasis.
- ☐ Bronchitis.
- ☐ Pneumonia.

54. On the basis of hypoxaemia: mild ARDS is characterized by.....

- ☐ $PaO_2/FiO_2 \leq 100$ mmHg.
- ☒ $PaO_2/FiO_2 > 200 \leq 300$ mmHg.
- ☐ $PaO_2/FiO_2 > 100 \leq 200$ mmHg.
- ☐ $PaO_2/FiO_2 > 300$ mmHg.

55. Which of the following laboratory abnormalities is common in sarcoidosis?

- ☒ Elevated serum angiotensin-converting enzyme.
- ☐ Presence of serum precipitins.
- ☐ Elevated Antineutrophil cytoplasmic antibodies.
- ☐ Elevated Anti-basement membrane antibodies.

56. Spirometry cannot be used to.....
 Differentiate obstructive from restrictive pulmonary disorders.
 Diagnosis and characterize severity of airway obstruction.
 Measure responses to therapy of obstructive airway disease.
 Diagnosis and characterize severity of pneumonia.

57. Iatrogenic pneumothorax refers to.....
 Pneumothorax in presence of underlying lung disease.
 Pneumothorax secondary to chest trauma.
 Pneumothorax that occurs spontaneously.
 Pneumothorax is secondary to thoracic procedure.

58. 29 years old male, nonsmoker, not known to have any chest diseases before. Presented by sudden mild chest pain and mild shortness of breath. On examination he is tall patients (189 cm), heart rate 88b/m, respiratory rate (22c/m), diminished breath sound over left side. X-ray showed pneumothorax occupying 20% of left hemithorax. Which of the following is the treatment of choice is.....

- Refer to cardiothoracic surgery.
- Oxygen therapy and observation.
- No intervention, just observation.
- Insert intercostal chest tube.

59. A 30-year-old, male, IV abuser presented with pleuritic chest pain and dyspnea. Patient gave history of cardiac catheterization. Chest X-ray showed multiple cavities surrounded by consolidation. Which one of the following is the most likely causative organisms?

- Staphylococcus aureus.
- Hemophilus influenza.
- Streptococcus pneumonia.
- TB.

60. Concerning acute lung abscess, which of the following is true?

- Less than two weeks.
- Less than three weeks. X
- Less than four weeks.
- Less than five weeks. X

2) SAQ
1) List 4 uses of opioid analgesics in cardiac patients?

.....
.....
.....
.....

(1 mark).

2) List four presentations of ischemic heart disease

.....
.....
.....
.....

(1mark)

3) Enumerate two causes of wide complex tachycard

.....
.....
.....
.....
.....
.....

4) List four factors affecting pulmonary blood pressure and flow

(1 mark).

.....
.....
.....
.....

5) Enumerate arterial supply of the lung?

(1 mark).

.....
.....
.....

6) Enumerate 4 diagnostic criteria of acute massive pulmonary embolism? (1 mark)

7) List 4 causes of exudative pleural effusion (1 mark).

8) Enumerate Etiology of myocardial infarction (1 mark).

3) EMQ

9) Select from column (A) the correct answer from column B
0.5 mark each)

Column A	Column B
1- Continuous and intermittent recorders are types of _____	a- Masked Hypertension
2- Elevated office BP and normal home BP _____	b- Heart Rate Variability
3- Balance between the cardiac sympathetic and vagal efferent activity _____	c- White coat hypertension
4- Normal office BP and elevated home BP _____	d- Ambulatory ECG monitoring

10) Each group of extended matching questions consists of lettered options followed by a list of numbered problems/questions. For each numbered problem/question select the one lettered option that most closely answers the question. You can use the lettered options once or not at all.
(2mark 0.5 mark each)

Options:

- A. Long term inhaled or oral antibiotic, chest physiotherapy and mucolytics
- B. Steroid and bronchodilators
- C. Mechanical ventilation and steroid
- D. Anticoagulant
- E. Chest tube insertion
- F. Antibiotics IV
- G. Chest phesotheraby and vaccinations
- H. Chemotherapy and radiotherapy

Problems/Questions

- 1. Lung abscess. **F**
- 2. Pulmonary embolism. **d**
- 3. Lung cancer. **H**
- 4. Bronchiectasis. **a**

4) MEQ

11) Compare between hypoxemic and hypercapnic respiratory failure (2 marks)

12) Enumerate Causes of pneumothorax? (2 marks)

5) CASE

13) Case 1

A 61-year-old, nonsmoker woman, presented by left sided dull aching agonizing chest pain, hemoptysis, and weight loss of about 10 kg during last 2 months. On exam heart rate 110 b/m, respiratory rate 30 c/m, blood pressure 110/78 mmHg, 2nd degree clubbing, and diminished breath sound over left side. X-ray showed massive left sided pleural effusion. Thoracentesis was done and its analysis revealed protein 6.9 gm, albumin 3.3 gm, LDH 499 IU, lymphocytes 85%. Serum analysis revealed protein 8.4 gm, albumin 4.2 gm, and LDH 550 IU. After thoracentesis CT chest revealed left lung mass. Bronchoscopy showed endobronchial mass in the left main bronchus. Bronchoscopic biopsy was taken and histopathology analysis revealed malignant cells while Immunohistochemistry revealed malignant cells positive for CEA, TTF, B72-3, CD5, and negative for calretinin and keratin CK5/6 (2 marks)

a. What is the most likely diagnosis for this case? (0.5 mark)

b. What is the type of this pleural effusion? (0.5 mark)

c. Mention TWO treatment modalities for this case (1 mark)

(2 marks)

14) Case 2
A 64-year-old female patient, bird breeder, resident at nearby factory. She was smoker but she had stopped smoking 6 months ago. She is complaining of dry cough, severe exertional dyspnea in the last one year and treating with aminopenicillins and ipratropium bromide + fenoterol for 7 days with minimal effect. Clinical examination revealed finger clubbing, tachycardia, and tachypnoea with multiple predominantly bibasal crackles, O₂ saturation was 85%, restrictive changes were identified at spirometry. Chest X-ray showed reticulonodular infiltration.

(0.5 mark)

- a. What is the most likely diagnosis of this case?

(0.5 mark)

- b. Mention 3 risk factors for this disease in the present case

(1 mark)

- c. Mention 2 confirmatory test for this condition

(2 marks)

15) Case 3
A 66-year-old female with a history of coronary artery disease and hypertension presents by ambulance to your emergency department, with acute progressive worsening of shortness of breath. She has noticed shortness of breath at night and has been resting on at least two pillows. Last night she was unable to sleep and had to stay in her recliner chair. She denies chest pain now or at any time this week. She admits that she failed to follow her low salt diet & missed some of her medications this week due to the holiday. The patient gave history recurrent attacks of gradual onset regular palpitation mainly with exertion and decrease by rest. The condition with associated swelling of both legs up to knee, with dyspepsia and pain in the right upper abdomen.

Physical exam:

- The patient is orthopneic and speaking in only three-word sentences.
- BP: 160/80 mmHg, Pulse: 105 b/min, RR: 32 c/min & Oxygen saturation: 89% on room air & 95% oxygen mask
- Neck examination: congested pulsating neck veins
- Chest auscultation: vesicular breathing with prolonged expiration with fine crackles in all lung fields.
- Cardiac auscultation: accentuated S₁ and S₂. An S₃ is present. There is no murmur.
- Abdominal examination: there is epigastria and right hypochondrial tenderness, no palpable organs or masses on exam.
- Lower limb: bilateral pitting edema to the level of knee

Form نموذج
B

What is the most probable diagnosis of this case and mention two important investigations to ensure diagnosis?

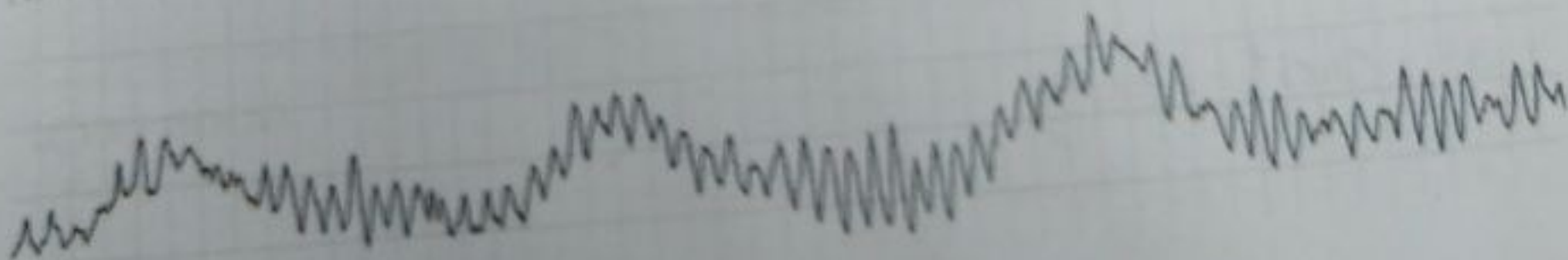
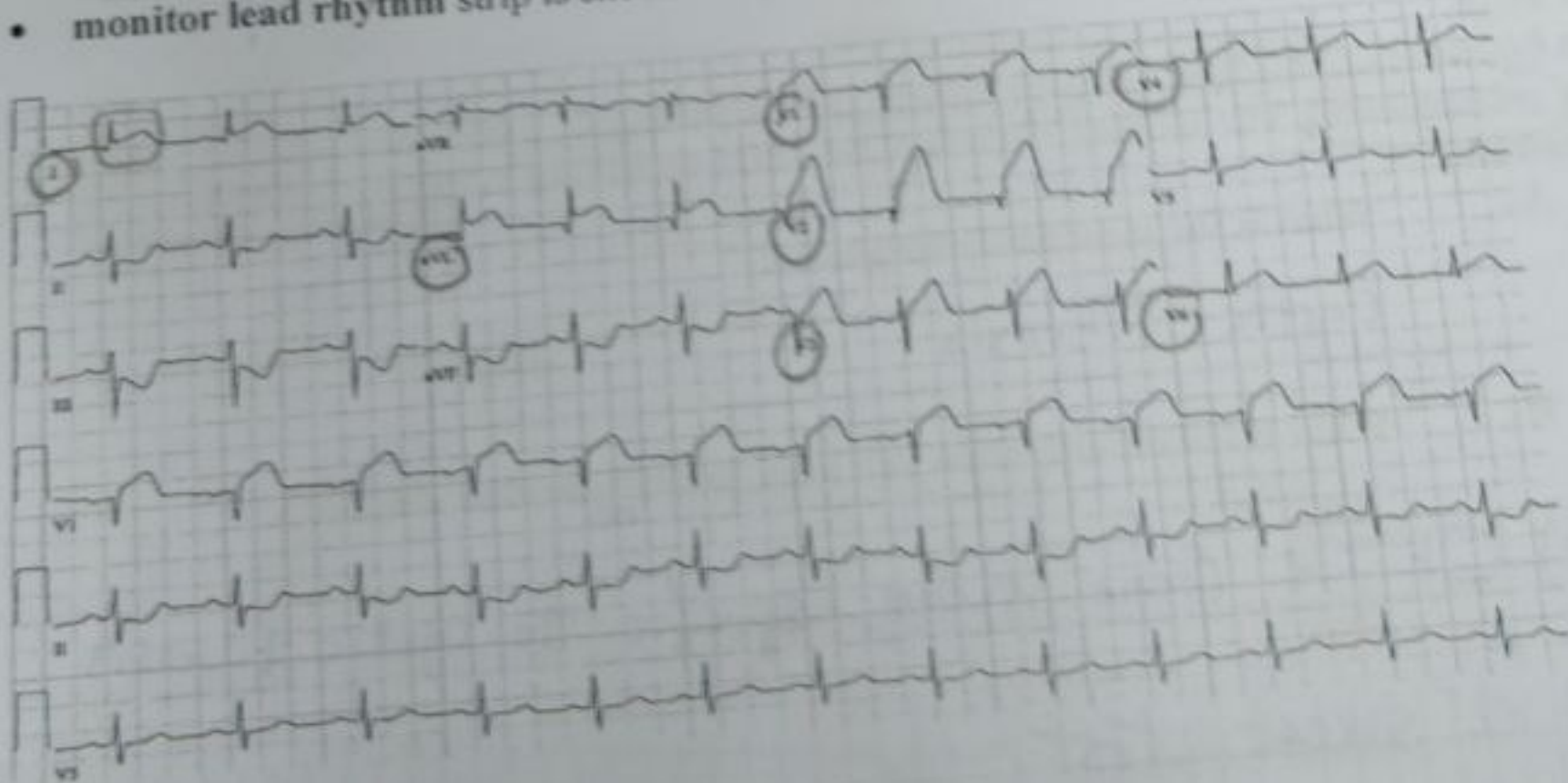
(2marks)

(6 marks)

16) Case 4

A 70-year-old gentleman presented to emergency department with progressive severe retrosternal compressing chest pain of two hours duration. The pain is associated with sweating, nausea, and vomiting. The patient is known to be a smoker with medical history of hypertension and type 2 diabetes mellitus on regular treatment. on examination: he was in severe pain, restless in bed, sweaty and dyspneic.

- Vital signs:
- Pulse: 80 bpm, regular and equal ✓
- BP: 160/100 mmHg equal in both side ↑
- Respiratory rate: 25 c/m with O2 sat. 92% ↓
- Chest auscultation: fine bilateral basal crepitation on the back ✓
- Cardiac auscultation: normal S1, accentuated A2 with audible S4
- ECG is done and soon after the patient became unresponsive.
- monitor lead rhythm strip is shown.



Q1) What is the explanation of previous ECG and monitor lead strip?

(2 mark)

ECG of Anteroseptal with high lateral wall STEMI and rhythm strip shows ventricular fibrillation.

ECG of Inferior wall STEMI and rhythm strip shows ventricular fibrillation.

ECG of Lateral wall STEMI and rhythm strip shows supraventricular tachycardia.

ECG of Anteroseptal with high lateral wall STEMI and rhythm strip shows atrial fibrillation.

(2 mark each, 1 per

Q2) list two main lines of treatment of this patient.
(point)

Q3) Enumerate two complications that could happen to this patient. (2 mark each, 1 per point)

17) Case 5

A 10-month-old infant has poor weight gain, a persistent cough, and a history of several bouts of pneumonia. The mother describes the child as having very large, foul-smelling stools for months.

A- What is the most likely diagnosis?

B- What is the investigation you need to confirm your diagnosis?

GOOD LUCK